

HEALTH-LAB — Production Datasheet

EoS Health · Models EOS-HL-001 (Base) / EOS-HL-002 (Ultra) · Revision 1.0

Document: EOS-DS-004-HL · **Date:** June 1, 2026 · **Status:** Production Release

1. Product Overview

HEALTH-LAB is a flexible biosensor patch that continuously measures sweat biomarkers from the skin surface. It is the world's first wearable to simultaneously measure glucose, lactate, cortisol, potassium, sodium, and pH in real time using aerosol-jet printed nano-electrodes and time-multiplexed amperometric/potentiometric sensing. The patch adheres to the upper arm or chest and transmits data wirelessly to the EoS Health Hub app.

2. Tier Comparison

Specification	Base (EOS-HL-001)	Ultra (EOS-HL-002)
Analytes	4 (glucose, lactate, Na ⁺ , pH)	7 (+ cortisol, K ⁺ , uric acid)
Wear time	7 days	14 days
Glucose accuracy	±15% (ISO 15197)	±5% (SCBN Kalman)
Iontophoresis	Passive sweat only	Active (carbachol stimulation)
MCU	nRF52833	nRF52840
Battery	30 mAh	65 mAh
Retail price	\$49/patch	\$89/patch
Subscription	None	None

3. Electrochemical Specifications

Analyte	Method	Electrode	Range	Accuracy	Tier
Glucose	Amperometric (GOx)	NEBA Pt nano	40–400 mg/dL	±15% (Base), ±5% (Ultra)	Both
Lactate	Amperometric (LOx)	NEBA Pt nano	0.5–20 mmol/L	±10%	Both
Sodium (Na ⁺)	Potentiometric (ISE)	NEBA Ag/AgCl	10–200 mmol/L	±5 mmol/L	Both
pH	Potentiometric	NEBA Ir/IrO ₂	4.0–8.0	±0.1 pH	Both
Cortisol	Aptamer-based	NEBA Au nano	1–200 ng/mL	±20%	Ultra
Potassium (K ⁺)	Potentiometric (ISE)	NEBA Ag/AgCl	1–50 mmol/L	±3 mmol/L	Ultra
Uric acid	Amperometric (UOx)	NEBA Pt nano	0.1–1.5 mmol/L	±15%	Ultra

NEBA: Nano-Electrode Bio-Array — aerosol-jet printed platinum/gold/iridium nano-electrodes on polyimide substrate.

4. Electrical Specifications

Parameter	Base	Ultra	Unit
Battery capacity	30	65	mAh
Active current (sensing + BLE)	3.8	5.2	mA
Iontophoresis current (Ultra only)	—	0.5	mA
Deep sleep current	2.1	2.8	μA
Battery life	7	14	days
Charging	NFC (13.56 MHz)	NFC (13.56 MHz)	—

5. Mechanical and Physical Specifications

Parameter	Value
Patch dimensions	55 × 35 × 2.5 mm
Weight	4.2 g
Substrate	Polyimide (Kapton) + medical-grade silicone adhesive
Electrode material	Aerosol-jet printed Pt/Au/Ir nanoparticles on polyimide
Adhesive	3M 2477P medical-grade acrylic adhesive
Skin contact biocompatibility	ISO 10993-5 (cytotoxicity), ISO 10993-10 (sensitization)
IP rating	IPX7 (1 m, 30 min)
Bend radius	≥5 mm (flexible substrate)
Wear locations	Upper arm, chest, lower back
Single-use	Yes (patch body is single-use; electronics module is reusable)

6. System Architecture

HEALTH-LAB uses a two-part design:

Disposable patch body: Contains the NEBA electrodes, reference electrode, iontophoresis electrodes (Ultra), and adhesive layer. Single-use, replaced every 7 or 14 days.

Reusable electronics module: Snaps onto the patch body via a 12-pin ZIF connector. Contains the MCU, BLE radio, battery, and analog front-end. Reused across multiple patch bodies.

7. Processor and Memory

Component	Base	Ultra
MCU	nRF52833	nRF52840
AFE	LMP91000 (3-channel)	LMP91002 (7-channel)
NVM	W25Q16JV (2 MB)	W25Q32JV (4 MB)
Secure element	ATECC608B	ATECC608B

8. SCBN Kalman Calibration (Ultra only)

The HEALTH-LAB Ultra uses a 3-reference electrode Kalman filter (SCBN — Self-Calibrating Biosensor Network) that continuously corrects for electrode drift, temperature variation, and sweat pH interference:

Parameter	Value
Kalman state vector	[glucose, drift_rate, pH_correction, temp_correction]
Process noise covariance	$Q = \text{diag}([0.01, 0.001, 0.005, 0.002])$
Measurement noise covariance	$R = 0.04$ (2% sensor noise)
Calibration interval	Continuous (every 30 s)
Glucose accuracy improvement	$\pm 15\% \rightarrow \pm 5\%$ vs. uncalibrated

9. BOM Summary (Production, 1k units)

Category	Base Cost	Ultra Cost
MCU + AFE	\$3.80	\$6.40
NEBA electrodes (aerosol-jet print)	\$4.20	\$8.60
Polyimide substrate + adhesive	\$2.10	\$3.40
Power (PMIC + battery + NFC)	\$3.20	\$5.80
Electronics module housing	\$2.80	\$3.60
Assembly + calibration	\$2.40	\$3.80
Packaging (3-pack)	\$1.80	\$2.40
Total BOM (per patch)	\$20.30	\$34.00
Target retail	\$49	\$89

10. Sweat Biomarker Reference Ranges

Biomarker	Normal Range	Clinical Significance
Glucose	40–120 mg/dL (fasting)	Diabetes monitoring, hypoglycemia alert
Lactate	0.5–2.0 mmol/L (rest)	Exercise intensity, metabolic stress
Sodium	10–90 mmol/L	Hydration status, cystic fibrosis screening
pH	4.5–7.5	Skin health, metabolic alkalosis
Cortisol	5–25 ng/mL (morning)	Stress, adrenal function, sleep quality
Potassium	5–25 mmol/L	Electrolyte balance, cardiac risk
Uric acid	0.2–0.8 mmol/L	Gout risk, purine metabolism